Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Work, Energy & Power

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. A force of 20 N moves a particle 5 m in the direction of the force. Find the work done. **[2]**

2. A particle of mass 3 kg is lifted vertically through 4 m. Find the work done against gravity. **[3]**

3. A particle of mass 2 kg is moving at $5\,\text{m}\,\text{s}^{-1}$. Find its kinetic energy. **[2]**

4. A particle of mass 4 kg falls freely through 10 m. Find the gain in kinetic energy. **[3]**

5. A force of 15 N acts at 30° to the direction of motion. The particle moves 6 m. Find the work done by the force. **[3]**

6. A car of mass 1000 kg accelerates from rest to $20\,\text{m}\,\text{s}^{-1}$. Find the work done by the engine (ignore resistance). **[3]**

7. A particle of mass 5 kg slides down a smooth plane through a vertical height of 3 m. Find its speed at the bottom if it starts from rest. **[4]**

8. A particle of mass 2 kg is projected up a smooth plane with speed $8\,\text{m}\,\text{s}^{-1}$ and rises through a vertical height of 2 m. Find its speed at that point. **[4]**

9. A machine does 500 J of work in 10 s. Find its power. **[2]**

10. A car engine produces a constant driving force of 800 N while travelling at $15\,\text{m}\,\text{s}^{-1}$. Find the power. **[3]**

11. A particle of mass 3 kg has speed $4\,\text{m s}^{-1}$ at the bottom of a smooth slope and speed $2\,\text{m s}^{-1}$ at height h. Find h. **[4]**

12. A force of 25 N moves a particle 8 m against a resistance of 5 N. Find the work done by the force and the work done against resistance. **[4]**

13. A particle of mass 6 kg is lifted at constant speed through 5 m by a crane. Find the work done by the crane. **[3]**

14. A particle of mass 4 kg moving at $10\,\text{m s}^{-1}$ is brought to rest by a constant resistance. Find the work done by the resistance. **[3]**

15. State the principle of conservation of mechanical energy. **[2]**

Section B – Hard (Q16–25)

16. A particle of mass 5 kg slides down a rough plane through a vertical height of 4 m. Its speed at the bottom is $6\,\text{m s}^{-1}$ having started from rest. Find the work done against friction. **[5]**

17. A car of mass 1200 kg travels up a slope of inclination $\sin^{-1}(1/20)$ at constant speed $15\,\text{m s}^{-1}$. The resistance is 200 N. Find the power of the engine. **[6]**

18. A particle of mass 2 kg is projected with speed $10\,\text{m s}^{-1}$ up a rough plane inclined at 30° ($\mu = 0.2$). Find how far up the plane it travels. **[6]**

19. A pump raises 50 kg of water per second through a height of 8 m and delivers it at $4\,\text{m s}^{-1}$. Find the power of the pump. **[5]**

20. A particle of mass 3 kg moves from rest under a constant force of 12 N. Find its kinetic energy after travelling 5 m. **[3]**

21. A car of mass 800 kg accelerates from $10\,\mathrm{m\,s^{-1}}$ to $25\,\mathrm{m\,s^{-1}}$ over 100 m against a constant resistance of 150 N. Find the work done by the engine. **[5]**

22. A particle of mass m slides from rest down a smooth curved path, dropping a vertical height h. Show that its speed at the bottom is $\sqrt{2gh}$. **[3]**

23. A lorry of mass 5000 kg is travelling at $20\,\mathrm{m\,s^{-1}}$ on a horizontal road. The engine is switched off and the lorry comes to rest after 80 m. Find the average resistance. **[4]**

24. A particle of mass 4 kg is pulled up a rough plane inclined at 20° ($\mu = 0.25$) at constant speed $3\,\mathrm{m\,s^{-1}}$ by a force parallel to the plane. Find the power. **[6]**

25. A pump takes water from a lake and delivers it at $6\,\mathrm{m\,s^{-1}}$ to a height of 12 m at a rate of $20\,\mathrm{kg\,s^{-1}}$. Find the power output of the pump. **[5]**

End of Worksheet • *Show all working clearly. Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless otherwise stated.*